TCL

BlueArk X10

C&I Energy Storage System



MULTI-ENERGY INTEGRATION

- Supports PV and diesel generator inputs for grid-connected, off-grid and weak-grid applications, including islands, industrial parks and microgrids.
- Grid-tied/off-grid transfer time <20 ms, ensuring uninterrupted power supply to critical loads.
- ESS - Genset Transfer Time < 20 ms in Both Standalone and Parallel Operation, Ensuring Continuous Off-Grid Power for Critical Loads

HIGH PV & STORAGE COMPATIBILITY

- 1500 V high-voltage platform with an MPPT voltage range of 300 - 1500 V, compatible with high-voltage modules and long-string designs.
- Up to 8 MPPTs, with 2 strings per MPPT for flexible string configuration.

STRONG OFF-GRID CAPABILITY

- Up to 125 kVA off-grid output to meet backup power requirements for critical loads.
- 1.1 × rated power continuous output in off-grid mode; up to 1.55 × rated power for short-duration overload.
- Supports black start for autonomous system startup even without grid or PV power.

FLEXIBLE DEPLOYMENT

- All-in-one integrated design, factory pre-integrated for rapid on-site deployment.
- Supports multi-unit parallel operation for easy capacity expansion.
- Centralized multi-site management helps reduce overall O&M labor costs.

Model	X10-125K-261
DC Input	
Max. PV Input Power [Wp] @ STC	250000
Max. Input Voltage [V]	1500
MPPT Voltage Range [V]	300–1500
Rated Input Voltage [V]	1050
Startup Voltage [V]	400
Max. Input Current [A]	8 × 40
Max. Short-Circuit Current [A]	8 × 55
No. of PV Strings per MPPT	2
Battery Parameters	
Battery Type	LiFePO4
Rated Energy [kWh]	261
Battery Voltage Range [V]	728–936
Max. Charge/Discharge Current [A]	179
AC Output (Grid Side)	
Rated Output Power [W]	125000
Max. Output Power [VA]	125000
Rated Output Current [A]	180.4
Max. Output Current [A]	189.9
Rated Voltage [V]	3L + N + PE, 380 V / 400 V
Rated Frequency/Range [Hz]	50/60
Power Factor Adjustment Range	-0.8 to +0.8
Total Harmonic Distortion [THDi]	<3%
AC Input (Grid)	
Rated Input Voltage [V]	3L + N + PE, 380 V / 400 V
Rated Input Frequency [Hz]	50/60
Max. Input Current [A] (Off-Grid Loads Disabled)	189.9
Max. Input Current [A] (Off-Grid Loads Enabled)	379.8
AC Output (Off-Grid)	
Max. Output Power [VA]	125000
Peak Output Power [VA]	1.1 × rated power, continuous; 1.55 × rated power, 1 s
Rated Voltage [V]	3L + N + PE, 380 V / 400 V
Rated Frequency/Range [Hz]	50/60
Total Harmonic Distortion [THDv] (Linear Load)	<3%
AC Input (Generator)【Built-in ATS】	
Max. Input Power [kW]	250000
Max. Input Current [A]	379.8
Rated Voltage [V]	3L + N + PE, 380 V / 400 V
Rated Frequency/Range [Hz]	50 / 60
Transfer Time	< 20ms
Protection	
PV Reverse Polarity Protection	Integrated
Anti-Islanding Protection	Integrated
AC Overcurrent Protection	Integrated
AC Short-Circuit Protection	Integrated
AC Overvoltage Protection	Integrated
DC Switch	Integrated
DC Surge Protection	Type II
AC Surge Protection	Type II
Arc-Fault Detection	Integrated
Rapid Shutdown	Optional
System Parameters	
Communication	Ethernet / RS485 / 4G
Topology	Transformerless
Operating Temperature Range	-30°C to +50°C (derating from 45°C to 50°C)
Cooling Method	Liquid cooling (battery) / Smart air cooling (inverter)
Operating Humidity	0–95%, non-condensing
Max. Operating Altitude	3,000 m
Ingress Protection	Battery compartment: IP55; Inverter: IP66
Dimensions	W 1420 × D 1350 × H 2355 mm
Weight	Approx. 2.45 t
Warranty [Years]	5